

EDUCATION

Bachelor of Engineering — Computer Science Engineering

Sathyabama Institute of Science and Technology, CGPA: 8.3 / 10
Nadu

Aug 2024– May 2028

Chennai, Tamil

INTERNSHIP EXPERIENCE

Artificial Intelligence Intern NoviTech R&D Private Limited | Jul 2024 – Aug 2024 | Certificate ID: AIIN3859

- Built a **conversational AI chatbot** using Python with NLP-driven intent recognition and dynamic response generation, delivering end-to-end dialogue management functionality
- Designed and evaluated **deep learning classification models** using TensorFlow and scikit-learn, applying feature engineering pipelines with NumPy and Pandas to improve model accuracy and training efficiency.
- Tracked and versioned model experiments using MLflow, enabling reproducible training runs and systematic hyperparameter comparison
- Managed source code and collaborated on version control workflows using **Git** and **GitHub**, following **CI/CD** best practices

PROJECTS

Predictive Maintenance System (2026) | Python · PySpark · XGBoost · Power BI

- Architected an end-to-end ML pipeline ingesting real-time industrial IoT sensor data to predict equipment failures **before occurrence**, reducing potential downtime risks proactively.
- Applied **Python, PySpark, and Apache Spark** for large-scale time-series sensor data preprocessing and distributed computing.
- Engineered domain-specific features and tuned **XGBoost** with Bayesian optimization, achieving **99.66% ROC-AUC**, outperforming baseline models.
- Leveraged **SHAP** for model explainability, identifying **Torque and Tool Wear** as top predictors of machine failure.
- Designed an interactive **Power BI dashboard** to visualize machine health, failure type distribution, and model performance metrics.

AWS Cloud Data Pipeline (2026) | AWS S3 · Lambda · RDS · boto3 · CloudWatch

- Designed a **serverless, event-driven pipeline** automating CSV ingestion, transformation, and reporting, improving scalability and reducing manual effort.
- Applied **AWS S3** for data ingestion, **Lambda (Python + boto3)** for transformation, and **RDS PostgreSQL** for structured storage.
- Configured **CloudWatch alarms** for real-time monitoring and automated recovery workflows.
- Implemented **IAM least-privilege policies** and cost optimization strategies for secure, efficient deployment.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, C++, Java, Bash/Shell Scripting.
- AI & Machine Learning:** TensorFlow, PyTorch, XGBoost, MLflow, Model Evaluation.
- Data Engineering & Big Data:** Apache Spark, PySpark, Apache Kafka, ETL/ELT Pipelines.
- Cloud Platforms:** AWS — S3, Lambda, RDS, boto3.
- Databases & Data Warehousing:** PostgreSQL, MySQL, MongoDB.
- MLOps & DevOps:** Docker, Kubernetes, Git, CI/CD, Linux/Unix.
- Data Visualization & Analytics:** Power BI, Microsoft Excel (Advanced), Data Analysis.

CERTIFICATIONS

Certification	Issuer	Year
Python (Basic)	HackerRank	2026
Python for Data Science Course	Reliance Foundation	2025
AI-DevOps Engineering Foundation	Reliance Foundation	2026
Python for Data Science (Elite)	NPTEL	2025

ACHIEVEMENTS & AWARDS

Luminary Award — Excellence in Academics Sathyabama Institute of Science and Technology | 2024–2025

- Awarded institutional recognition for outstanding academic performance and dedication during the first year of BE Computer Science Engineering.

2nd Prize – HacktheHorizon 2025 (24-hour Hackathon) | VIT Chennai

- Secured 2nd place in a competitive 24-hour hackathon, developing an innovative solution under time constraints and demonstrating strong technical and collaborative skills.